Questions

Q1.

Figure 15 shows a foam jacket around a copper cylinder.

Figure 15

The hot water is stored in the copper cylinder until it is needed.

The foam jacket helps to keep the water hot.

(i) Explain how the foam helps to keep the water hot.

(2)

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* (ii) A company has developed a new material which they think could be used instead of foam around the cylinder.

Devise an investigation they could carry out to make a fair comparison of the insulating properties of their new material with those of the foam.

(6)

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(Total for question = 8 marks)

Q2.

Figure 2 shows a solar farm.

(Source: © Science Photo Library)

Figure 2

The solar panels use energy from the Sun to generate electricity.

Electricity generated in this way is sometimes called 'free' electricity.

(i) Give **one** reason why this might be called 'free' electricity.

(1)

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(ii) Give **one** reason why this electricity is **not** 'free'.

(1)

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(iii) In one second, 1000 J of energy from the Sun reaches 1 m^2 of the surface of a solar panel.

In one second, the amount of energy from the Sun reaching the whole surface of a solar panel is 2800 J.
Calculate the area of this solar panel.

(2)

area = ... m^2

(Total for question = 4 marks)

Q3.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 16.

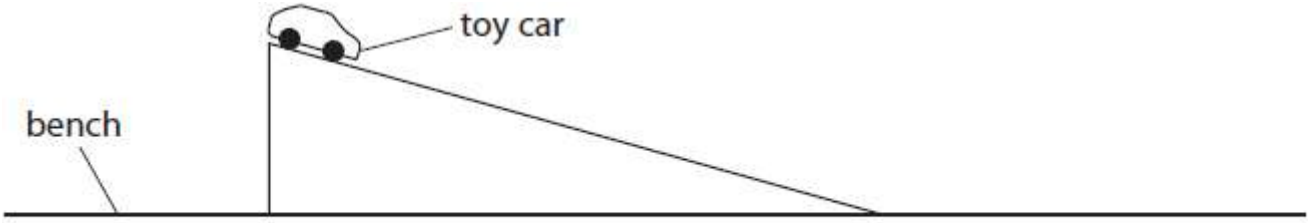

Figure 16

Explain **one** way the student could reduce the amount of thermal energy transferred to the surroundings as the toy car rolls down the slope.

(2)

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(Total for question = 2 marks)

Q4.

In the UK, electricity is generated using non-renewable and renewable energy resources.

The graph in Figure 18 shows how the amount of electricity generated by these resources changed from 2012 to 2020.

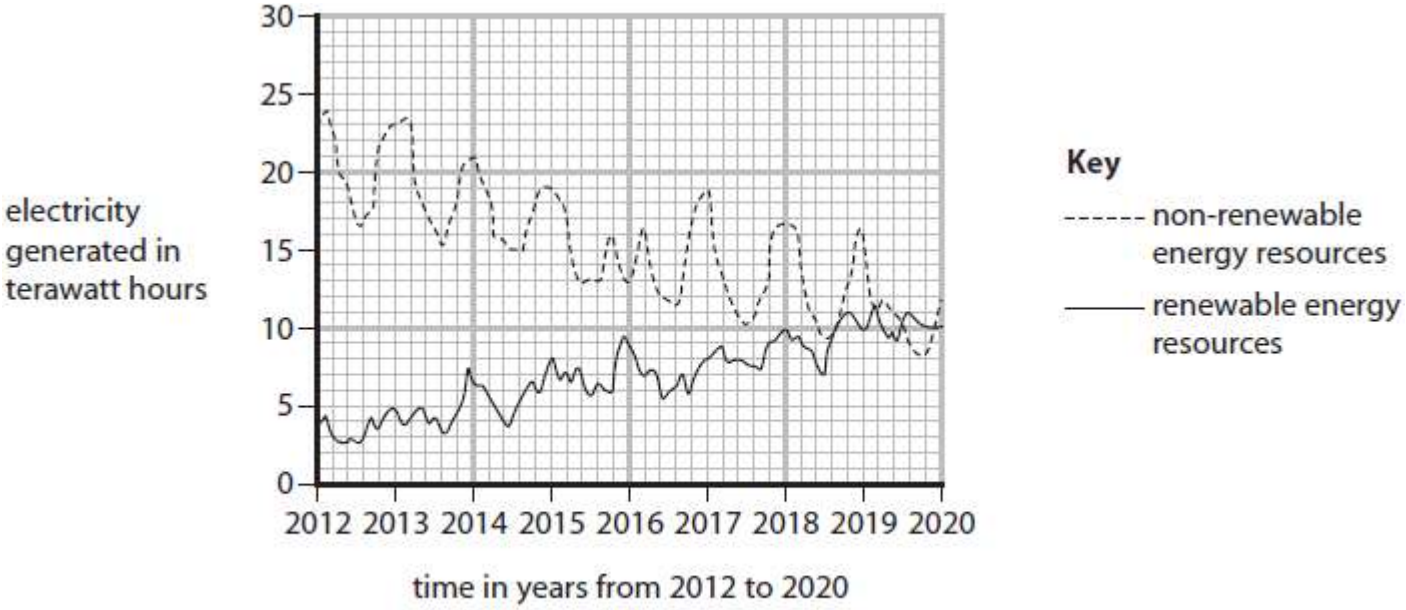

Figure 18

Explain how and why the amount of electricity generated by renewable and nonrenewable energy resources has changed from 2012 to 2020.

Your answer should include

- the trends shown in Figure 18
- the change in the amount of electricity generated by at least one renewable resource
- the change in the amount of electricity generated by at least one non-renewable resource.

(6)

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(Total for question = 6 marks)

Q5.

Figure 10 shows wind power generation from the year 2000 onwards.

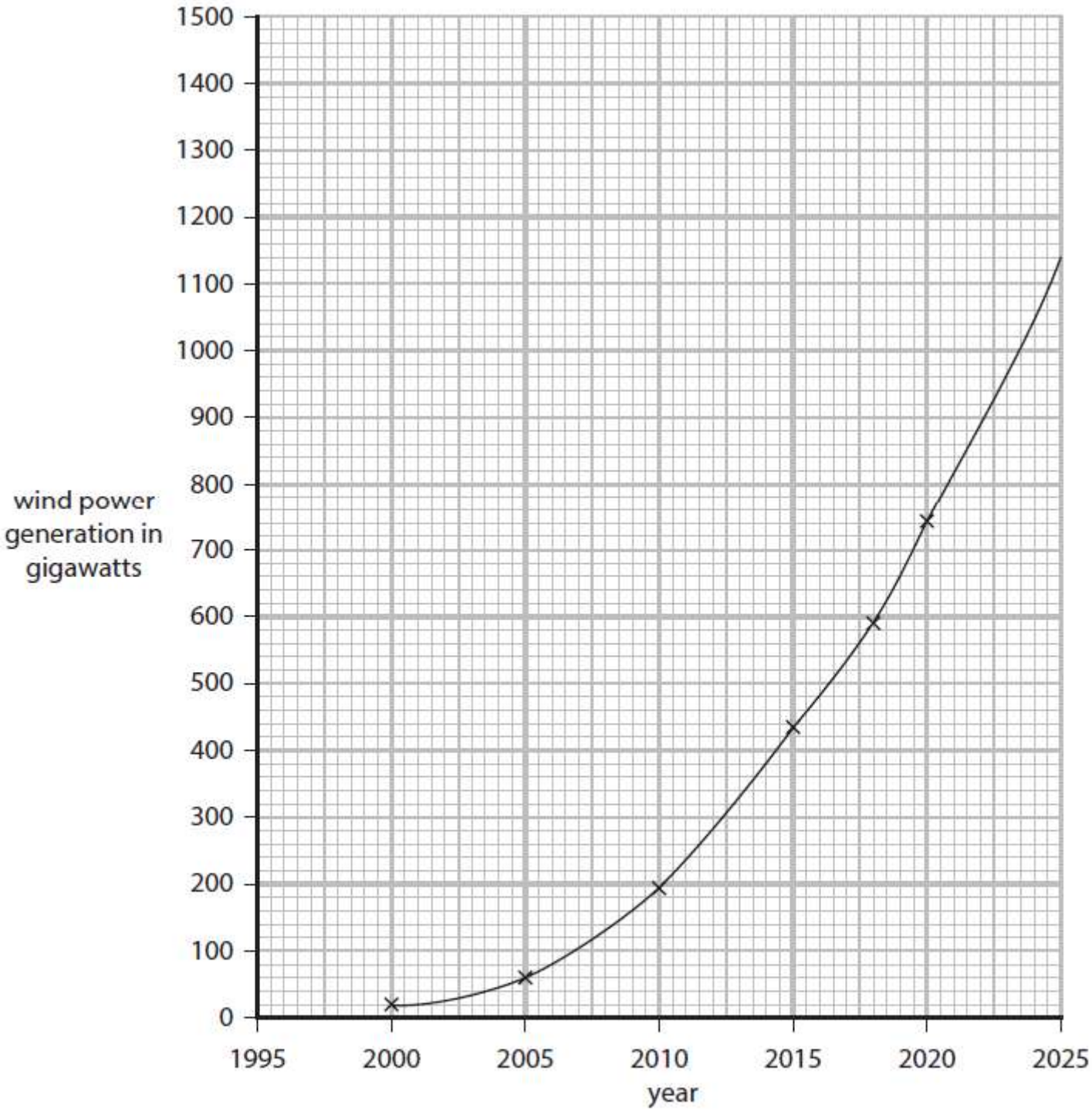

Figure 10

Estimate the percentage change in wind power generation from 2005 to 2020.

(3)

percentage change = ... %

(Total for question = 3 marks)

Q6.

This photograph shows a fan.

The blades of the fan are turned by an electric motor.

In one second, the motor gets 200 J of electrical energy from the mains supply.
Only 180 J of this energy is used to turn the blades of the fan.

The rest of the energy is wasted.

(i) Calculate how much of the 200 J of energy is wasted.

(1)

wasted energy = ... J

(ii) State what happens to the wasted energy.

(1)

.....

.....

(iii) Calculate the efficiency of the motor.

(2)

efficiency = ...

Q7.

The photographs show two ways of supplying thermal energy.

Use words from the box to complete the sentence under each photograph.

chemical	electrical	kinetic	light	sound

(i) The photograph shows a kettle.

The kettle transfers ..... energy to thermal energy.

(1)

(ii) The photograph shows a barbecue.

The barbecue transfers ..... energy to thermal energy.

(1)

Q8.

Figure 6 shows a spinning flywheel.

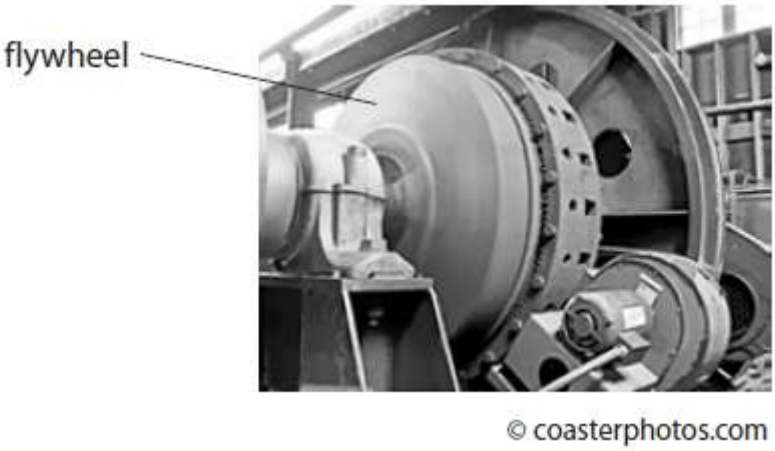

Figure 6

(i) State how energy is stored in a spinning flywheel.

(1)

.....

(ii) State **one** way to increase the amount of energy stored in the flywheel.

(1)

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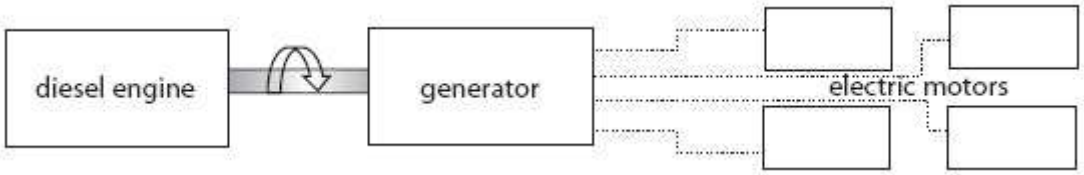
(Total for question = 2 marks)

Q9.

(a) A train is powered by a diesel engine.
The diesel engine is used to turn a generator.
The generator provides electricity for electric motors which drive the wheels.



(i) Draw one straight line from each train part to its useful energy transfer.

(3)

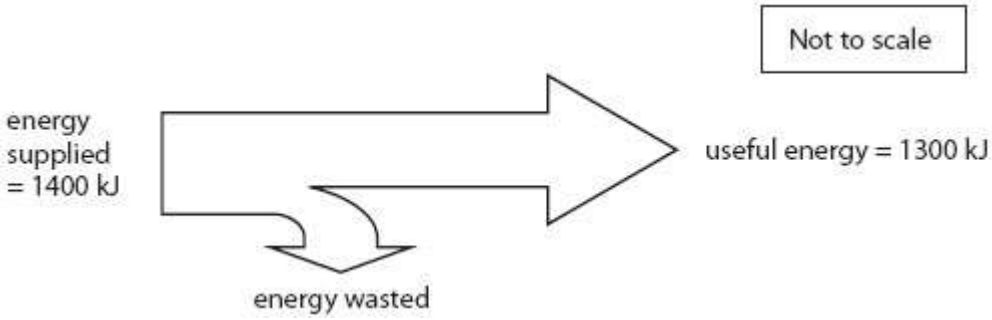
train part	useful energy transfer
diesel engine	chemical to electrical
generator	chemical to kinetic
motor	electrical to kinetic
	kinetic to chemical
	kinetic to electrical

(ii) State **one** example of a non-useful energy transfer in the motor.

(1)

.....

(b) The diagram represents the energy transfer in one second in the generator.



(i) Calculate the amount of energy wasted in one second in the generator.

(1)

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(ii) Calculate the efficiency of the generator.

(2)

.....

(c) The electric motors which drive the wheels are painted black.
Suggest why the motors are painted black.

(1)

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(Total for Question is 8 marks)

Q10.

A student uses the apparatus in Figure 12 to find out which of two materials, sand or sawdust, is the better insulator.

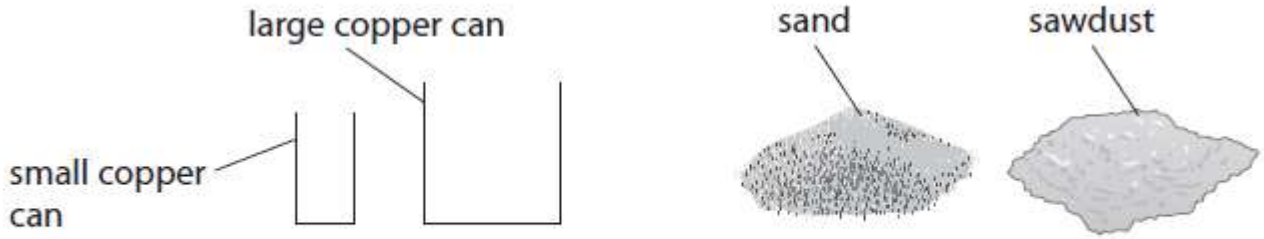

Figure 12

The student also has a kettle to boil water, a thermometer and a stop clock.

(i) Draw a labelled diagram to show how the student should set up the equipment to investigate which material is the better insulator.

(3)

(ii) Give **three** factors that the student must control in this investigation.

(3)

1 .....

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2 .....

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3 .....

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(Total for question = 6 marks)

Q11.

Expanded polystyrene, used to insulate buildings, has different densities.

Figure 13 shows how the thermal conductivity of expanded polystyrene changes with the density of expanded polystyrene.

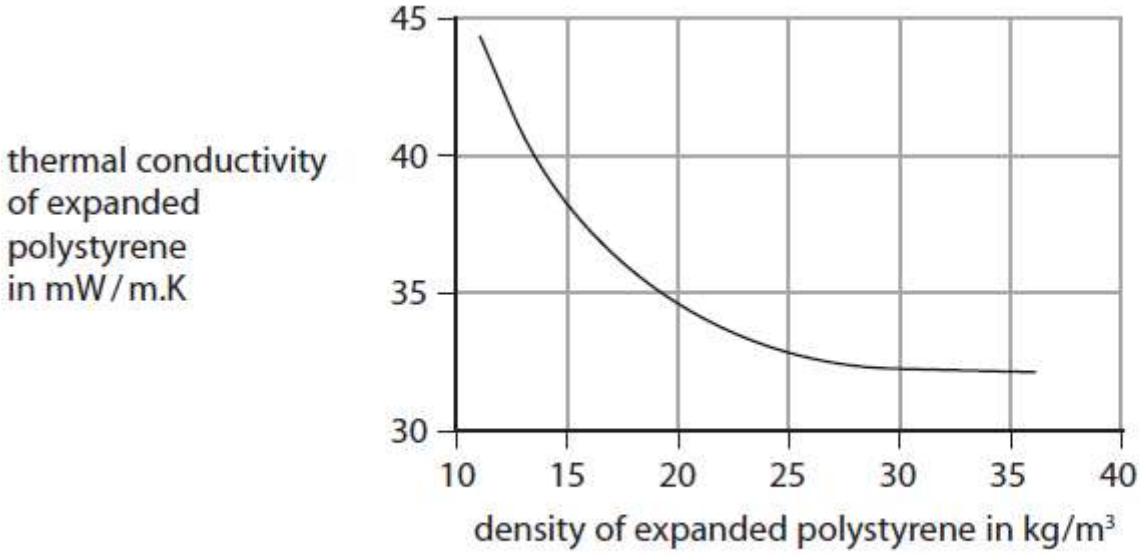

Figure 13

Using the graph in Figure 13, describe how the thermal conductivity of expanded polystyrene changes with the density of expanded polystyrene.

(2)

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(Total for question = 2 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	An explanation that combines identification - understanding (1 mark) and reasoning/justification - understanding (1 mark): foam contains pockets of air OR foam / air is an insulator (1) which reduces energy transfer to surroundings (1)		(2)

Question number	Indicative content	Mark
(ii)	Answers will be credited according to the candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **A03 (6 marks)** <u>Apparatus</u> • beakers • thermometer(s) • stop watch • foam and new material <u>Measurement of energy loss</u> • put material around cylinder/ beaker/calorimeter (or use foam jacket to start with) • put hot water into cylinder/beaker • measure temperature • measure temperature at intervals/at start / at finish/after fixed period of time • or measure time taken to cool to set temperature <u>Comparison between two materials</u> • change to other material around cylinder/beaker • compare the temperatures of the two after fixed time • or compare time taken to cool • the better insulator cools down more slowly <u>Measures to ensure fair test</u> • same mass of water • same thickness of material • same starting temperature • same time interval • same room temperature <u>Accuracy</u> • repeat readings • take average of readings	(6)

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Analyses the scientific information but understanding and connections are flawed. (AO3) • An incomplete plan that provides limited synthesis of understanding. (AO3)
Level 2	3–4	• Analyses the scientific information and provides some logical connections between scientific enquiry, techniques and procedures. (AO3) • A partially completed plan that synthesises mostly relevant understanding, but not entirely coherently. (AO3)
Level 3	5–6	• Analyses the scientific information and provide logical connections between scientific enquiry, techniques and procedures. (AO3) • A well-developed plan that synthesises relevant understanding coherently. (AO3)

Q2.

Question number	Answer	Additional guidance	Marks
(i)	• idea that the source of energy is the sun (which is free) (1)	ignore 'renewable'	**(1)**

Question number	Answer	Additional guidance	Marks
(ii)	One from: • the setup / installing of the solar panels is not free (1) • cost of materials involved (1) • cost of maintenance (1)	ignore 'expensive' by itself – must be qualified ignore 'pay electricity bills'	**(1)**

Question number	Answer	Additional guidance	Marks
(iii)	substitution (1) 2800 ÷ 1000 answer (1) 2.8 (m^2)	allow rounded to 3 (m^2) (no power of ten error allowed here) award full marks for correct numerical answer without working	**(2)**

Q3.

Question number	Answer	Additional guidance	Mark
	An explanation linking: lubricate/oil the wheels (1) (to) reduce friction (1)	 make the toy car more streamlined / different surface / lubricate slope reduce air resistance / drag accept start from lower down the slope (1) (to) reduce the total amount of energy (transfer) (1)	**(2)**

Q4.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO2,AO3** **Non-renewable sources of energy** trend: less used/decrease in use (between 2012 and 2019) **fossil fuels** coal, gas, oil are running out / finite resource / sustainability argument produce carbon dioxide/ sulphur dioxide/ greenhouse gases (when burned) in power stations cause pollution/ smoke particles /damage to the environment causes climate change / global warming production of greenhouse gases needs to be reduced (for Britain to become carbon neutral)	**(6)** **AO2, AO3**

	nuclear fuels no carbon dioxide produced radioactive waste produced safety concerns **Renewable sources of energy** trend: more used /increase in use (between 2012 and 2019) renewable and non-renewable about equally used from 2019 solar, wind, hydroelectric, tidal, geothermal, wave and biomass never run out / are sustainable do not produce carbon dioxide/ greenhouse gases (except biomass) slow down climate change / global warming	

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts about the resources, non-renewable or renewable **OR** the trend(s) in usage	Possible candidate responses coal is non-renewable and solar is renewable non-renewables are decreasing and renewables are increasing non-renewable resources are higher on (most of) the graph
Level 2	3–4	Additional guidance trend(s) **AND** limited explanation of the renewable trend **OR** limited explanation of the non-renewable trend	Possible candidate responses use of renewable resources is increasing because renewables are sustainable OR use of non-renewable resources are decreasing because they cause global warming

Level 3	5–6	Additional guidance both trends **AND** detailed explanation of one trend **AND** some explanation of the other trend	Possible candidate responses use of renewable resources are increasing **and** the use of non-renewable resources are decreasing because non-renewable resources are running out **and** wind turbines do not produce carbon dioxide

Q5.

Q6.

Question Number	Answer	Acceptable answers	Mark
(i)	20 (J)	200 – 180 (even if calculated value from this is incorrect)	**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	(changed to) {thermal energy / heat}	dissipated (lost) to {surroundings / motor / air / atmosphere} sound / noise **reject** if kinetic, light or chemical is mentioned	**(1)**

Question Number	Answer	Acceptable answers	Mark
(iii)	$\frac{180}{200} \times 100$ (1) 90 (%) (1)	award full marks for correct answer with no working $\frac{180}{200}$ 0.9, 9/10 Or [100 – (20/200)] % not needed but if a unit is given then maximum score is 1	**(2)**

Q7.

Question Number	Answer	Acceptable answers	Mark
(i)	electrical	electric	**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	chemical		**(1)**

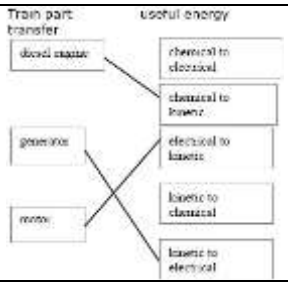
Q8.

Question Number	**Answer**	**Additional guidance**	**Mark**
(i)	kinetic (1)	**only** (adding another incorrect alternative negates)	**(1)** AO 2 1

Question Number	**Answer**	**Additional guidance**	**Mark**
(ii)	any one of increase the speed (of spinning) (1) increase the mass / weight (of the flywheel) (1)	accept (idea of) faster ignore make it bigger	**(1)** AO 2 1

Q9.

	Answer	**Acceptable answers**	**Mark**
(ai)	A line connecting a train part with a useful energy transfer as shown below (1)	Lines need not be straight Ignore any arrow heads drawn Note: if more than one line is drawn from a train part then zero mark for that train part.	**(3)**

			
(aii)	(transfer of energy to) thermal (1)	heat/sound	**(1)**
(bi)	1400 – 1300 (= 100) (kJ) (1)		**(1)**
(bii)	Substitution (1) 1300 / 1400 × 100 Evaluation (1) 93(%) or 0.93	A value which rounds to 93(%) or 0.93 Correct answer with no working scores 2 marks	**(2)**
(c)	Any one from black is a good thermal radiator (1) (helps to) prevent motors overheating (1)	(good) emitter (helps to) remove wasted energy/ heat (from the motor)	**(1)**

Q10.

Question Number	Answer	Additional guidance	Mark
(ii)	any **three** factors from: {mass / volume} of water (1) {volume / thickness / mass} of insulators /materials (1) {starting / initial} temperature of water (1) time interval / temperature change (1)	 accept amount / specified values / "how much" accept amount / specified values / "how much" accept temperature of hot / boiling water / specified values accept specified values of interval or change unqualified "same time" is insufficient	**(3)** **AO3**

Q11.

	Answer	Additional guidance	Mark
	a description giving as the density (of expanded polystyrene) increases the (thermal) conductivity decreases (1) non-linear / gradient decreases / at a decreasing rate / levels off / plateaus / becomes (almost) constant (1)	 ORA allow inversely proportional / exponential for non-linear in this context ignore negative correlation unqualified quoted values are insufficient	(2) AO3